

# Quantum Feature Engineering for Credit Default Prediction: When and Why IQP Circuits Help Linear Classifiers

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**Abstract**—Credit default prediction is a tabular classification problem in which modest gains in  $F_1$  translate directly into reduced financial exposure. We ask whether *Instantaneous Quantum Polynomial-time* (IQP) circuits can produce features that improve a classifier over both its raw classical baseline and Kernel PCA—the strongest unsupervised classical non-linear alternative—at an equal feature budget. The dataset provides 23 financial attributes per client; for an  $n$ -qubit circuit we select  $n$  of them, encode each as a rotation angle, and read  $2n$  expectation values back out as new features. The motivation for using a quantum circuit is computational: an  $n$ -qubit IQP circuit runs in constant depth and encodes feature correlations in a  $2^n$ -dimensional Hilbert space, whereas classical simulation of its exact output statistics scales exponentially in  $n$ . Using the UCI *Default of Credit Card Clients* dataset and five-fold cross-validation, we find that appending 16 IQP features ( $n = 8$  qubits) to a Logistic Regression model raises  $F_1$  from 0.462 to 0.517 ( $+0.055$ ,  $p < 0.0001$ ). Kernel PCA, the next-best method, reaches only 0.493 at the same feature count; the gap survives Benjamini–Hochberg correction across 12 tests ( $p = 0.00007$ ). No other classifier—Random Forest, SVM, XGBoost, or k-NN—benefits, which points to a linear-expressivity mechanism rather than a generic improvement. We also show that *how* the 8 input features are chosen matters: Random Forest importance-guided selection reaches  $F_1 = 0.523$ , while encoding maximally uncorrelated features drops it to 0.496, demonstrating that the circuit amplifies informative structure rather than creating it from scratch.

**Index Terms**—quantum feature engineering, credit default prediction, IQP circuits, logistic regression, quantum-classical hybrid, financial machine learning

## I. INTRODUCTION

Credit card default prediction sits at the intersection of class imbalance, non-linear feature interactions, and regulatory pressure to use interpretable models. The UCI *Default of Credit Card Clients* dataset describes each client with 23 heterogeneous features—credit limits, demographics, and six months of payment history—and only  $\sim 22.6\%$  of clients default, so it is a realistic stress test for any feature-engineering strategy.

Logistic Regression is the workhorse of credit scoring, but its limitation is structural: it fits a single hyperplane in the original 23-dimensional feature space, so payment-history variables that interact *multiplicatively* cannot be separated by a straight decision boundary. Quantum feature maps address this by embedding the data into a high-dimensional Hilbert space in which those interactions are represented implicitly

through entanglement [1], [2]. Concretely, a parametric circuit maps a real input vector into a state in a Hilbert space of dimension  $2^n$ , and the expectation values of local observables on the output state become new features that encode non-linear combinations of the inputs. For IQP circuits specifically, the argument for eventually using real quantum hardware is not only expressivity but *cost*: the circuit runs in constant depth for any number of qubits  $n$ , whereas classically simulating its exact output distribution requires time exponential in  $n$  [3]–[5].

This paper studies the question empirically and on a classical simulator; that is, we ask whether IQP-derived features, *however they are computed*, improve a downstream linear classifier over the best classical alternative. Whether quantum hardware could compute these same features more cheaply than classical simulation is a separate, complexity-theoretic matter that we return to in Section V-C. Two concrete questions drive the work. First, do IQP features improve over the raw 23-feature baseline when appended to a Logistic Regression model? Second, do they beat Kernel PCA (KPCA) at an equal feature budget? KPCA is the appropriate *target of comparison* because it is, like the quantum map, unsupervised, non-linear, and produces abstract features without ever looking at the class label. Beating KPCA would mean the quantum map captures structure that the best classical unsupervised method misses.

Prior work on quantum machine learning for finance tends to test on small datasets, skip multiple-comparison corrections, or compare quantum against a weak baseline rather than the best available classical alternative [6]. We address all three issues. Our contributions are: (i) a nine-way equal-budget comparison of quantum and classical feature-extraction methods on a standard financial dataset; (ii) evidence that the quantum advantage is specific to linear classifiers and absent for Random Forest, SVM, XGBoost, and k-NN; (iii) FDR-corrected significance testing over 12 planned comparisons; and (iv) a study of five feature-selection strategies showing that RF-importance-guided selection outperforms arbitrary choice and that encoding uncorrelated features actively hurts performance.

## II. BACKGROUND AND RELATED WORK

### A. Quantum Feature Maps

A quantum feature map  $\phi : \mathbb{R}^d \rightarrow \mathcal{H}$  embeds a classical vector  $\mathbf{x}$  into a quantum state  $|\phi(\mathbf{x})\rangle$  living in a Hilbert space  $\mathcal{H}$  of dimension  $2^n$ , where  $n$  is the number of qubits. To turn that state back into usable numbers we measure a fixed set of  $K$  Hermitian *observables*  $\{O_k\}_{k=1}^K$  (in this work the per-qubit Pauli  $X$  and  $Y$  operators, so  $K = 2n$ ); each one yields a scalar feature

$$f_k(\mathbf{x}) = \langle \phi(\mathbf{x}) | O_k | \phi(\mathbf{x}) \rangle, \quad k = 1, \dots, K. \quad (1)$$

The index  $k$  simply enumerates the observable (equivalently, the output feature). The corresponding kernel  $\kappa(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2$  can represent decision boundaries that standard kernels express only with a very large number of terms—for example boundaries that depend on the *parity* or simultaneous sign of several features (XOR-like interactions), which a polynomial or RBF kernel approximates poorly at low order [1].

### B. IQP Circuits

IQP (*Instantaneous Quantum Polynomial-time*) circuits interleave Hadamard layers with diagonal commuting unitaries. Their output distributions are believed to be classically hard to sample [3], [4], and on quantum hardware their depth is *independent* of the number of qubits, which makes them attractive for near-term devices. Coyle et al. [5] propose IQP circuits as practical feature extractors precisely because expectation values are easy to estimate by sampling even when exact simulation is hard. The general feature-map formalism of the previous subsection applies directly to IQP; the next section makes the specific circuit, its inputs, and its outputs fully concrete.

### C. Credit Risk Modeling

Logistic Regression has dominated credit scoring for decades because regulators can inspect its coefficients and auditors can reconstruct individual decisions [7]. Kernel SVMs and gradient boosting have shown competitive accuracy [8], but remain black boxes that are difficult to certify under frameworks such as the EU’s model-risk guidelines. Quantum-enhanced approaches for credit modelling are nascent; most published results use toy datasets or do not compare against strong classical baselines.

## III. METHODOLOGY

### A. Dataset

We use the UCI *Default of Credit Card Clients* dataset (OpenML #42477) [9], using all  $N = 30,000$  clients for every experiment. The original input is a vector of 23 features per client (credit limits, demographic variables, and six monthly payment-history records); the positive (default) rate is 22.6%, creating a mild class imbalance. All experiments use five-fold stratified cross-validation, and reported metrics are the mean  $\pm$  standard deviation over folds.

### B. IQP Quantum Feature Extraction

This section is the core of the method, so we describe it end to end: what enters the circuit, what the circuit does, and what comes out. Figure 1 summarises the four-layer circuit, and the text below follows the same order.

a) *What enters the circuit (the input)*: The circuit has  $n$  qubits; throughout the main experiments  $n = 8$ . Because 8 qubits can accept only 8 angles, we first *select* 8 of the 23 classical features (the selection strategy is itself studied in Section IV-E). Each selected feature is standardised with a `QuantileTransformer` (mapping it to a normal distribution), clipped to  $[-3\sigma, 3\sigma]$ , and converted to an encoding angle

$$\theta_i = x_i \cdot \pi/3, \quad i = 1, \dots, n, \quad (2)$$

which places every angle in  $(-\pi, \pi)$  and so uses the full dynamic range of the rotation gates. The vector  $\boldsymbol{\theta} = (\theta_1, \dots, \theta_8)$  is the actual input to the circuit: one angle per qubit.

b) *What the circuit does (the four layers)*: The 8-qubit IQP unitary  $U_{\text{IQP}}(\boldsymbol{\theta})$  is built from four layers, shown left to right in Fig. 1:

- 1) **Superposition layer.** A Hadamard gate  $H$  is applied to every qubit, taking the all-zero state into a uniform superposition over all  $2^8 = 256$  computational basis states:

$$|0\rangle^{\otimes 8} \xrightarrow{H^{\otimes 8}} |+\rangle^{\otimes 8} = \frac{1}{\sqrt{2^8}} \sum_{x \in \{0,1\}^8} |x\rangle. \quad (3)$$

- 2) **Data encoding.** A single-qubit  $R_Z(\theta_i)$  rotation imprints feature  $\theta_i$  as a phase on qubit  $i$ :

$$R_Z(\theta_i) = \begin{pmatrix} e^{-i\theta_i/2} & 0 \\ 0 & e^{i\theta_i/2} \end{pmatrix}. \quad (4)$$

The input is now stored in the *phase* of the state, not its amplitude.

- 3) **Entanglement layer.** Two-qubit ISINGZZ gates are applied on a ring (qubit  $i$  coupled to qubit  $i+1 \bmod 8$ ):

$$\text{ZZ}_{ij}(\phi_{ij}) = e^{-i\phi_{ij} Z_i \otimes Z_j / 2}, \quad \phi_{ij} = \frac{1}{2}(\theta_i + \theta_j). \quad (5)$$

This layer is the source of the quantum advantage: it *entangles* neighbouring qubits, encoding pairwise (and, through the ring, higher-order) interactions between features in a way that has no efficient classical description as  $n$  grows.

- 4) **Interference layer.** A second Hadamard layer converts the accumulated phase information into amplitude differences, making the encoded interactions readable by measurement:

$$U_{\text{IQP}}(\boldsymbol{\theta}) = H^{\otimes 8} \left( \prod_{\langle i,j \rangle} \text{ZZ}_{ij}(\phi_{ij}) \right) \left( \prod_i R_Z(\theta_i) \right) H^{\otimes 8}. \quad (6)$$

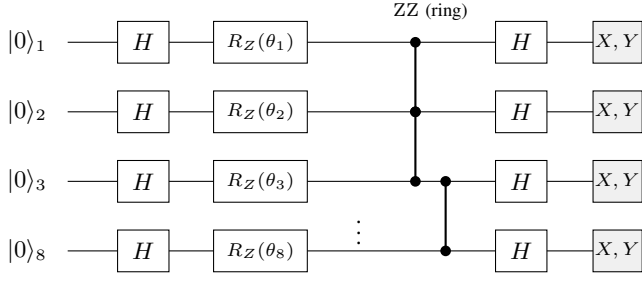


Fig. 1. The 8-qubit IQP feature extractor (four of the eight wires shown). Left to right: a Hadamard superposition layer, an  $R_Z(\theta_i)$  data-encoding layer, an ISINGZZ entanglement layer on a ring topology, and a second Hadamard interference layer. Each qubit is finally measured in the Pauli  $X$  and  $Y$  bases, producing  $2n = 16$  output features.

c) *What comes out (the output features):* After applying  $U_{\text{IQP}}$  we measure, for each qubit  $i$ , the expectation values of the Pauli  $X$  and  $Y$  observables:

$$f_{X,i} = \langle U X_i U^\dagger \rangle, \quad f_{Y,i} = \langle U Y_i U^\dagger \rangle. \quad (7)$$

This yields  $8 \times 2 = 16$  quantum features per client. (The Pauli  $Z$  expectations are identically zero for IQP circuits and are therefore excluded, which is why we read  $2n$  rather than  $3n$  features.) Each output feature is a non-linear function of *all* the input angles  $\theta_i$  that reflects inter-qubit entanglement; it cannot be factored into independent per-qubit contributions.

d) *Combined representation:* The 16 quantum features are concatenated with the original 23 classical features, giving a 39-dimensional input vector that is then fed to the downstream classifier. In other words, the quantum circuit is used as a feature *augmenter*, not a replacement.

e) *Encoding variants:* We test two encodings. The IQP encoding above uses the ZZ layer to couple qubit phases. As a simpler control, *Angle encoding* replaces the entanglement layer with a plain  $R_Y(\theta_i)$  rotation per qubit (followed by Hadamard and  $\langle X \rangle, \langle Y \rangle$  measurements), producing 16 features with *no* inter-qubit coupling. Both encodings are evaluated across all five classifiers, yielding 10 encoding-classifier pairs; each pair is also compared directly against KPCA, adding 2 more tests. This gives the family size  $m = 12$  used in the multiple-comparison correction (Section IV-C).

### C. Classical Comparison Methods

Each classical method adds exactly 16 features, matching the quantum budget (39 total). We include a noise baseline (16 random Gaussian features) as a sanity check against dimensionality inflation, and a family of *linear* transformations—PCA, SVD, ICA, random projection, and feature agglomeration—to establish whether *any* unsupervised transformation helps a linear classifier at all. Polynomial degree-2 terms test explicit pairwise interactions. Kernel PCA with an RBF kernel (bandwidth  $\gamma = 1/(d \cdot \text{Var}(\mathbf{x}))$ , the `sklearn` default) is our *primary* classical comparator: like the quantum approach it is unsupervised, non-linear, and produces abstract features without access to the class label. Methods that use supervised feature selection (e.g. polynomial

TABLE I  
PERFORMANCE WITH AND WITHOUT 16 IQP FEATURES, PER CLASSIFIER. “+QUANTUM” USES THE 8-QUBIT IQP ENCODING. VALUES ARE MEAN  $\pm$  STD OVER 5 FOLDS. ONLY LOGISTIC REGRESSION IMPROVES SIGNIFICANTLY.

| Classifier | Method    | $F_1$                              | Acc. (%)                        | AUC                                |
|------------|-----------|------------------------------------|---------------------------------|------------------------------------|
| Log. Reg.  | Classical | $0.462 \pm .014$                   | $67.5 \pm .3$                   | $0.708 \pm .014$                   |
|            | +Quantum  | <b><math>0.517 \pm .017</math></b> | <b><math>75.9 \pm .9</math></b> | <b><math>0.744 \pm .014</math></b> |
| Rand. For. | Classical | $0.500 \pm .026$                   | $79.9 \pm .5$                   | $0.753 \pm .010$                   |
|            | +Quantum  | $0.494 \pm .024$                   | $79.9 \pm .5$                   | $0.755 \pm .013$                   |
| SVM (RBF)  | Classical | $0.517 \pm .017$                   | $76.1 \pm .8$                   | $0.737 \pm .017$                   |
|            | +Quantum  | $0.519 \pm .018$                   | $76.1 \pm 1.1$                  | $0.745 \pm .016$                   |
| XGBoost    | Classical | $0.517 \pm .017$                   | $75.5 \pm 1.0$                  | $0.748 \pm .007$                   |
|            | +Quantum  | $0.509 \pm .027$                   | $76.4 \pm 1.7$                  | $0.739 \pm .017$                   |
| k-NN       | Classical | $0.396 \pm .009$                   | $77.9 \pm .3$                   | $0.683 \pm .014$                   |
|            | +Quantum  | $0.393 \pm .011$                   | $77.8 \pm .4$                   | $0.681 \pm .015$                   |

expansion followed by `SelectKBest`) have an inherent advantage and are reported for completeness only.

### D. Classifiers

We evaluate five classifiers: Logistic Regression (L2,  $C = 1.0$ , balanced weights), Random Forest (300 trees, balanced), SVM (RBF kernel), XGBoost (100 trees,  $lr = 0.1$ ), and k-NN ( $k = 5$ , distance-weighted). All use `sklearn` defaults except where stated.

### E. Statistical Testing

Significance is assessed with paired  $t$ -tests across the five CV folds ( $\alpha = 0.05$ ). To control for multiple comparisons we apply the Benjamini–Hochberg false discovery rate (FDR) procedure [10] over a family of 12 planned comparisons.

## IV. EXPERIMENTAL RESULTS

### A. Effect Across Classifiers

Table I reports each classifier with and without the 16 appended IQP features; values are mean  $\pm$  standard deviation over the five folds. The pattern is stark: Logistic Regression gains 0.055  $F_1$  and 8.4 percentage points of accuracy (both  $p < 0.0001$ ), while every non-linear classifier is statistically unchanged—Random Forest, SVM, XGBoost, and k-NN all return  $p > 0.05$  on every metric, and several change by less than one standard deviation.

The reason is structural. A linear classifier cannot form non-linear decision boundaries on its own, so the quantum features supply the non-linearity externally. Ensemble and kernel methods already construct such boundaries internally—through tree splits, an RBF kernel, or boosting residuals—so the same features carry redundant information for them. This also rules out a generic dimensionality-inflation effect: if merely adding 16 extra inputs helped, we would see gains across all five classifiers, which we do not.

TABLE II

LOGISTIC REGRESSION WITH 16 AUGMENTED FEATURES (39 TOTAL). ALL METHODS ARE UNSUPERVISED EXCEPT POLYNOMIAL<sub>16</sub> (SEE TEXT). **BOLD** MARKS THE BEST RESULT IN THE TABLE. PRIMARY METRIC IS  $F_1$ ;  $p$ -VALUES ARE FROM A PAIRED  $t$ -TEST VS. THE BASELINE.

| Method                      | $F_1$        | $\Delta F_1$  | Acc.          | $p$ -value         |
|-----------------------------|--------------|---------------|---------------|--------------------|
| Baseline (23)               | 0.462        | —             | 67.5 %        | —                  |
| Noise control               | 0.456        | −0.006        | 67.1 %        | 0.24               |
| PCA <sub>16</sub>           | 0.463        | +0.001        | 67.5 %        | 0.68               |
| SVD <sub>16</sub>           | 0.463        | +0.001        | 67.5 %        | 0.68               |
| ICA <sub>16</sub>           | 0.462        | +0.000        | 67.5 %        | 1.00               |
| RandProj <sub>16</sub>      | 0.463        | +0.001        | 67.5 %        | 0.73               |
| FeatAgg <sub>16</sub>       | 0.462        | −0.001        | 67.5 %        | 0.37               |
| Poly <sub>16</sub>          | 0.470        | +0.008        | 69.1 %        | 0.16               |
| KPCA <sub>16</sub>          | 0.493        | +0.031        | 73.1 %        | 0.0001             |
| <b>Quantum<sub>8q</sub></b> | <b>0.517</b> | <b>+0.055</b> | <b>75.9 %</b> | <b>&lt; 0.0001</b> |

### B. Nine-Way Comparison at Equal Feature Budget

Table II ranks all nine feature-engineering methods at 39 features each for Logistic Regression; in every case the *same* 16 extra features are computed once and reused inside each cross-validation fold (no per-fold refitting of the transform on test data). Quantum features place first. The five linear transformations—PCA, SVD, ICA, random projection, and feature agglomeration—all leave  $F_1$  essentially unchanged ( $\Delta \leq 0.001$ ,  $p > 0.37$ ). This is expected rather than surprising: a *linear* transformation of features fed to a *linear* model stays inside the same hypothesis class, so it cannot change the set of achievable decision boundaries—the linear methods serve as a control that confirms only genuinely non-linear maps can help here. Polynomial degree-2 terms add 0.008  $F_1$ , a real but modest gain. KPCA does substantially better, adding 0.031 ( $p = 0.0001$ ), confirming that non-linearity is what matters. Quantum features add 0.055, roughly 1.8 $\times$  the KPCA gain ( $p = 0.00007$  for the quantum–KPCA gap). The random-noise baseline is the one method that *hurts*  $F_1$  slightly, ruling out any explanation based purely on the increase in dimensionality.

### C. Multiple-Comparison Correction

With 12 pre-specified tests—10 encoding–classifier pairs plus two direct quantum-versus-KPCA comparisons—at  $\alpha = 0.05$  we would expect roughly one false positive by chance alone. Applying Benjamini–Hochberg FDR correction (Table III) leaves three findings intact: IQP outperforms both the baseline and KPCA for Logistic Regression, and Angle encoding outperforms the baseline. Angle encoding does *not* survive when compared directly to KPCA ( $p \approx 0.12$ ), making IQP the only encoding with a defensible claim to genuinely exceed the best classical unsupervised alternative. All eight tests involving non-linear classifiers remain non-significant.

### D. Quantum Features as Supplement, Not Replacement

To test whether quantum features can stand alone, we ran three IQP circuits in parallel—8, 8, and 7 qubits respectively—so that every one of the 23 input features was encoded in at least one circuit. Each qubit contributes a Pauli  $X$  and Pauli  $Y$  measurement, giving  $8 \times 2 + 8 \times 2 + 7 \times 2 = 46$  quantum

TABLE III

BENJAMINI–HOCHBERG FDR CORRECTION ( $m = 12$ ,  $\alpha = 0.05$ ). PRIMARY METRIC:  $F_1$ . ONLY RESULTS WITH RANK  $\leq 3$  SURVIVE.

| Rank | Hypothesis                                | $p$ -value     | BH crit. | Survives |
|------|-------------------------------------------|----------------|----------|----------|
| 1    | IQP vs. KPCA (LogReg)                     | 0.00007        | 0.0042   | ✓        |
| 2    | IQP vs. Baseline (LogReg)                 | < 0.0001       | 0.0083   | ✓        |
| 3    | Angle vs. Baseline (LogReg)               | < 0.0001       | 0.0125   | ✓        |
| 4    | Angle vs. KPCA (LogReg)                   | $\approx 0.12$ | 0.0167   | ×        |
| 5–12 | Quantum vs. Baseline (RF, SVM, XGB, k-NN) | $\approx 0.5$  | —        | ×        |

TABLE IV

$F_1$  BY FEATURE-SELECTION STRATEGY (LOGISTIC REGRESSION, 5-FOLD CV). STRATEGIES S1–S4 APPEND QUANTUM FEATURES TO ALL 23 CLASSICAL FEATURES (39 TOTAL); S5 USES 3 CIRCUITS + 23 CLASSICAL (69 TOTAL). **BOLD** = BEST  $F_1$ .

| Strategy                   | Selection method      | $F_1$        | Acc.          | AUC          |
|----------------------------|-----------------------|--------------|---------------|--------------|
| Baseline (no quantum)      | —                     | 0.462        | 67.5 %        | 0.708        |
| S1: First-8 (arbitrary)    | Features 1–8          | 0.517        | 75.9 %        | 0.744        |
| <b>S2: RF Importance</b>   | <b>Top-8 by Gini</b>  | <b>0.523</b> | <b>75.3 %</b> | <b>0.748</b> |
| S3: Correlation-based      | Top-8 corr. w/ target | 0.516        | 75.5 %        | —            |
| S4: Diverse (uncorrelated) | Greedy min-corr       | 0.496        | 72.6 %        | —            |
| S5: Multi-circuit + orig.  | All 23 (3 circuits)   | 0.519        | 74.6 %        | —            |

features with complete feature coverage and no classical input. Accuracy was 69.4% and  $F_1 = 0.485$ , well below the 75.9% and 0.517 achieved when the same circuits are combined with the original 23 features. Full coverage does not rescue quantum-only prediction: the model needs the raw financial variables, not just their quantum projections.

### E. Feature Selection Strategy Comparison

An 8-qubit circuit encodes exactly 8 of the 23 available features, so the choice of which 8 to use is a real design decision. Table IV compares five strategies.

The ordering in Table IV is not random. RF importance (S2) is a deterministic, data-driven procedure: a Random Forest is trained on the full 23 features and features are ranked by mean decrease in Gini impurity. The same 8 features are selected every time on the same data, unlike the arbitrary first-8 (S1), which simply takes whatever order the dataset happens to use. The selected features are primarily payment-history and credit-limit variables ( $\{f_5, f_6, f_7, f_0, f_{11}, f_{12}, f_{17}, f_{18}\}$ ), and the result,  $F_1 = 0.523$ , is the highest across all strategies—that is, it slightly exceeds even the strong 0.517 obtained by the arbitrary first-8. Correlation-based selection (S3) uses similar logic and scores 0.516, close but not identical. The arbitrary first-8 (S1) lands in between at 0.517; it happens to include several moderately predictive features, which explains why it is competitive despite no deliberate selection.

The most instructive result is the diverse strategy (S4), which greedily selects features with the lowest mutual correlation.  $F_1$  drops to 0.496—below every other strategy and statistically worse than first-8 ( $p < 0.01$ ). Maximally uncorrelated features spread the circuit’s encoding capacity across the full variable space, but most of those variables carry little signal about default. The IQP circuit does not generate predictive

power; it amplifies interactions among already-informative inputs. When those inputs are weak, the entanglement layer has nothing useful to couple.

Covering all 23 features with three separate 8-qubit circuits (S5) raises the feature count to 69 and yields  $F_1 = 0.519$ , better than arbitrary first-8 but still below RF importance and at considerably greater cost. In practice, selecting the top-8 features by any reasonable ranking method is both simpler and more effective.

## V. DISCUSSION

### A. Why Only Logistic Regression?

The quantum kernel induced by the IQP circuit,  $\kappa(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2$ , defines a decision boundary in a high-dimensional Hilbert space. For Logistic Regression—a linear model in feature space—appending quantum features is equivalent to lifting the input into a richer representation where a linear boundary can capture non-linear structure in the original space. Random Forest, SVM with an RBF kernel, and XGBoost already implement their own non-linear transformations through tree splits, the RBF kernel matrix, or boosting residuals. The IQP features add no information they cannot already express, so the gain is zero.

### B. Feature Importance of Quantum Features

In the 39-feature logistic regression model, the four largest absolute coefficients all belong to quantum features:  $Q\_X1$  (0.766),  $Q\_X5$  (0.615),  $Q\_Y1$  (0.551), and  $Q\_Y6$  (0.474). Quantum features hold 10 of the top-20 coefficient positions despite making up only 41% of the feature set—meaning the model relies on them more heavily than their share of the input would predict. This is not an artefact of scaling or regularisation; it reflects that quantum features carry predictive information the classical variables, fed individually into a linear model, do not capture on their own.

### C. Limitations

All circuits were run on a classical simulator (PennyLane `default.qubit`). Our claims are therefore about the *predictive value* of IQP-derived features, not about a realised quantum speed-up: at  $n = 8$  qubits the statistics are trivially simulable, and the complexity-theoretic hardness only becomes relevant at much larger  $n$ . Gate noise and decoherence on real hardware would also alter the expectation values and likely narrow the gap over KPCA. Whether the advantage survives on near-term devices is an open question that should be settled before drawing hardware-deployment conclusions.

## VI. CONCLUSION

IQP feature extraction provides a statistically robust  $F_1$  improvement for Logistic Regression on credit default prediction: +0.055 over the raw baseline ( $p < 0.0001$ ) and +0.024 over the best unsupervised classical alternative, Kernel PCA ( $p = 0.00007$ , FDR-corrected over 12 tests). The same features produce no benefit for Random Forest, SVM, XGBoost, or

k-NN, which locates the mechanism firmly in the inability of linear models to construct non-linear boundaries on their own.

Feature selection for the quantum circuit is not a detail to be glossed over. Encoding the top-8 features by Random Forest importance is the most reliable strategy ( $F_1 = 0.523$ ), while encoding maximally uncorrelated features actively damages performance ( $F_1 = 0.496$ ). The circuit amplifies interactions among informative inputs; it cannot compensate for encoding variables that carry little signal about the outcome.

Future work should evaluate these circuits on real quantum hardware with error mitigation, investigate whether variational training of the circuit parameters can close the gap further, and examine the fairness properties of quantum-augmented credit scores under consumer-protection regulations.

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